

The Basel Problem

A story of an impossible sum, a young genius,
and a surprise visit from π

A Simple Question

What happens when you add up the reciprocals of the perfect squares?

$$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots$$

In modern notation:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = ?$$

The Challenge (1650)

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The sum converges (by comparison with $\int_1^\infty x^{-2} dx = 1$). The partial sums creep toward something near **1.6449...**

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“If anyone finds and communicates to us that which thusfar has eluded our efforts, great will be our gratitude.”

— Jakob Bernoulli, 1689

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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

π ? In a sum that has nothing to do with circles?

Step 1: Start with $\sin(x)$

We know the Taylor series for sine:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

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Note the coefficient of x^2 on the right: it is $-\frac{1}{6}$.

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Euler treated it like a polynomial and factored it by its roots:

$$\frac{\sin(x)}{x} = \left(1 - \frac{x^2}{\pi^2}\right) \left(1 - \frac{x^2}{4\pi^2}\right) \left(1 - \frac{x^2}{9\pi^2}\right) \cdots$$

Assumes: An entire function is determined by its zeros (up to normalization). Justified by the Weierstrass factorization theorem (1876).

Step 3a: How to Read the x^2 Coefficient from a Product

Consider just **three** factors — write $a_k = \frac{1}{k^2\pi^2}$ for short:

$$\begin{aligned} & (1 - a_1x^2)(1 - a_2x^2)(1 - a_3x^2) \\ &= 1 - (a_1 + a_2 + a_3)x^2 + (\text{terms in } x^4, x^6) \end{aligned}$$

Each x^2 contribution comes from picking $-a_kx^2$ from **one** factor and 1 from the rest. Cross-terms yield x^4 or higher.

So the x^2 coefficient is just $-(a_1 + a_2 + a_3)$.

Step 3b: Apply to the Infinite Product

$$\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2 \pi^2} \right) = 1 - \left(\sum_{n=1}^{\infty} \frac{1}{n^2 \pi^2} \right) x^2 + \dots$$

The x^2 coefficient is:

$$- \left(\frac{1}{\pi^2} + \frac{1}{4\pi^2} + \frac{1}{9\pi^2} + \dots \right) = -\frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Assumes: Absolute convergence and term-by-term expansion, as for finite products.

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Coefficients must agree: $\frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{6}$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Multiply both sides by π^2 :

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots = \frac{\pi^2}{6} \approx 1.6449 \dots$$

Why It Matters

Euler's proof was **not rigorous** by modern standards — the factoring step needed Weierstrass's theory of infinite products to justify, over 140 years later (1876).

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The sum we computed is $\zeta(2)$, a special value of the **Riemann zeta function**:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

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- Euler showed all **even** values $\zeta(2), \zeta(4), \zeta(6), \dots$ are rational multiples of powers of π
- But the **odd** values $\zeta(3), \zeta(5), \zeta(7), \dots$ remain mysterious — no closed forms are known

What We Learned

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

- A simple question can hide a profound answer
- The boldness to treat $\sin(x)/x$ as a polynomial was the key insight
- π connects geometry and number theory in ways we are still discovering

Sources

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